Predicting March Madness Basketball Games

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Outline

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2. Data
3. Past Strategies
4. Modeling Techniques
5. Methods

Section 1

Background

Introduction

- March Madness is the NCAA Division I Basketball Tournament, a single-elimination bracket of 68 teams held each March and April across both the Men's and Women's divisions to crown the National Champions.

- Teams earn a spot either by winning their conference tournament (an automatic bid) or by being selected by the NCAA Selection Committee (an at-large bid).

- The regular season determines the seeding. Teams are seeded 1 through 16 within each of four regions, where lower seeds are considered the strongest and are rewarded with matchups against the highest seeds.

- With 63 games decided by single elimination, upsets are frequent, which is why the tournament has become famous for its unpredictability, hence the name "March Madness."

Tournament Bracket

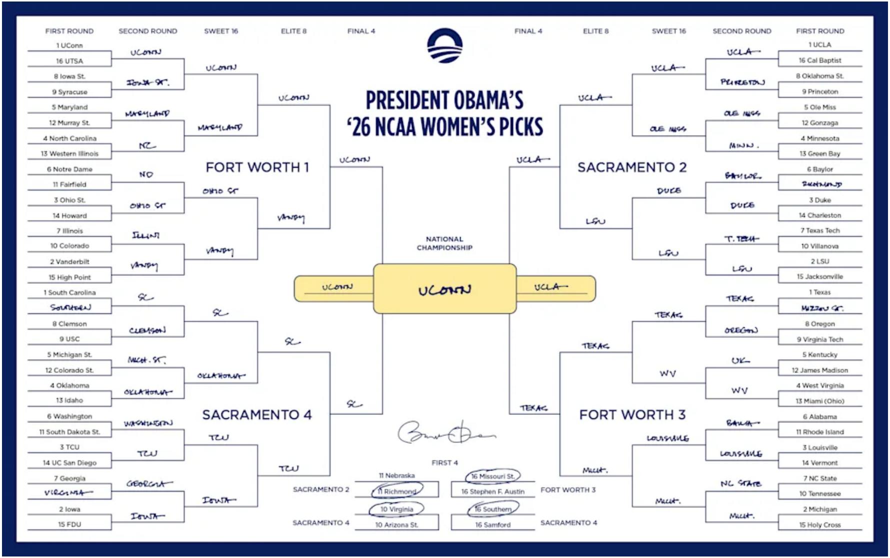

Figure 1: 2026 Bracket Example

Kaggle

- Kaggle is an online platform for data science and machine learning competitions, owned by Google.
- Companies, researchers, and organizations post prediction problems with datasets, evaluation metrics, and prize pools.
- Participants submit models that are scored on a hidden test set, with a public leaderboard ranking submissions in real time.
- Kaggle has become a benchmark in the ML community for evaluating predictive modeling approaches against a global field of competitors.

March Machine Learning Mania

- Annual Kaggle competition running since 2014, challenging participants to predict outcomes of every possible matchup in the NCAA tournament.

- Submissions are a probability $p \in [0, 1]$ for each of the 4556 possible matchups across Men's and Women's.

- Scored using the Brier Score:

$$BS = \frac{1}{N} \sum_{t=1}^{N} (f_t - o_t)^2 \tag{1}$$

- Where f_t is our prediction and o_t is the outcome.
- o_t is 1 if the team wins, and 0 is they lose.

Brier Score

- A naive 'coin flip' strategy (50% probability for all games) yields a Brier Score of 0.25, serving as our baseline.

Predicted	Outcome	Scenario	Brier Score
0.50	Win or Loss	Baseline	0.2500
1.00	Win	Perfectly confident, correct	0.0000
1.00	Loss	Perfectly confident, incorrect	1.0000
0.70	Win	Moderately confident, correct	0.0900
0.70	Loss	Moderately confident, incorrect	0.4900
0.90	Win	Highly confident, correct	0.0100
0.90	Loss	Highly confident, incorrect	0.8100

Table 1: Brier Score Examples

Prediction Difficulty

- Top predictive models plateau near 75% accuracy [**?**].
- Odds of a perfect bracket: 1 in 9.2 quintillion (random) to 1 in 128 billion (informed).
- No publicly verified bracket has survived past the Sweet Sixteen [**?**].
- $3.1 billion wagered in 2025, sportsbook lines are the benchmark to beat [**?**].

Section 2

Data

Kaggle Data

- Kaggle provides 37 tables spanning over three decades, separated by gender.
- Contains detailed regular season and playoff results, plus tournament seeding tables.
- Detailed box scores begin in 2003 for men and 2010 for women.

Variable	**Description**
PTS	Points scored
FGM / FGA	Field goals made and attempted
FGM3 / FGA3	Three-point field goals made and attempted
FTM / FTA	Free throws made and attempted
OR / DR	Offensive and defensive rebounds
Ast	Assists
TO	Turnovers
Blk	Blocks
PF	Personal fouls

Table 2: Detailed Result Variables

External Data Sources

- Box scores miss nuances such as pace of play and strength of schedule.
- Supplement Kaggle data with advanced efficiency metrics:
  - **KenPom** (Men's): industry standard, per-possession efficiency adjusted for opponent [**?**].
  - **Bart Torvik** (Women's): comparable methodology, used since KenPom is men's only [**?**].

Metric	Description
NetRtg	ORtg − DRtg
ORtg	Pts / 100 poss.
DRtg	Pts allowed / 100 poss.
AdjT	Possessions per 40 min

Table 3: KenPom Metrics

Metric	Description
ADJOE	Adjusted offensive efficiency
ADJDE	Adjusted defensive efficiency
AdjT	Adjusted tempo

Table 4: Bart Torvik Metrics

Data Considerations

- **2020 Tournament Gap:** the COVID-canceled tournament produces no playoff observations; regular season 2020 data is still usable.
- **Completeness:** no missing values across Kaggle, KenPom, or Torvik. Every team has a complete statistical profile.
- **Team Name Matching:** external sources use inconsistent spellings (ex: "St. Mary's" vs "Saint Mary's CA").
    - Stage 1: exact match on normalized names against Kaggle lookup
    - Stage 2: fuzzy match using Jaro-Winkler distance with threshold 0.3
    - All teams successfully matched in both divisions

Section 3

Past Strategies

Academic Foundation

- Tournament seeding alone is a strong predictor via logistic regression with z-score transformation [**?**].
- Combined logistic regression and Markov chain models, benchmarked against nearly 100 ranking systems, established logistic regression as the baseline standard [**?**].
- No classifier (decision trees, random forests, naive Bayes, MLPs) exceeds 74–75% accuracy on NCAAB data [**?**].
- **Key takeaway:** feature construction matters more than model choice.

Dominant Frameworks

- **Fundamental Analysis:** construct a "true" probability independent of public sentiment.
  - Elo ratings [**?**], KenPom, FiveThirtyEight, Massey
  - Treats expert outputs as features, not final predictions
- **Efficient Market Hypothesis** [**?**]: betting markets aggregate information better than any single model.
  - Uses closing lines from major bookmakers.
  - `goto_conversion` algorithm [**?**] corrects for favorite-long shot bias.

NIL Era and Recent Baselines

- **Boosting** [?]: tiered features (seeding to efficiency to Elo to GLM team quality), implemented in XGBoost. Multiple top 10 finishes in 2024 and 2025 cite his work as the baseline for their submission.
- **Simulation pipelines** among top 2024 finishers: predict point margins, then simulate brackets thousands of times via normal distribution over margins.
- Inspired the three models in this paper: **Logistic Regression**, **XGBoost**, and **Linear Regression Monte Carlo**.

Section 4

Modeling Techniques

Logistic Regression

- Binary classification model that applies the sigmoid function to a linear combination of features.

$$P(Y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k)}} \tag{2}$$

- $\mathbf{x} = (x_1, \ldots, x_k)$: feature vector
- $\boldsymbol{\beta} = (\beta_0, \ldots, \beta_k)$: coefficients estimated via maximum likelihood
- Output is bounded between $[0, 1]$, making it interpretable as a probability.
- Assumes linear relationship between features and log-odds; nonlinearity addressed via polynomial or interaction terms.

Linear Regression

- Models a continuous outcome as a linear combination of features, with coefficients estimated by ordinary least squares [**?**].

$$\hat{Y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k \tag{3}$$

- Unlike logistic regression, output is unbounded and not a probability.
- In sports prediction, used to estimate point margins which are then converted to win probabilities via:
  - logistic transformation of the predicted margin, or
  - Monte Carlo simulation over the margin distribution.

Visualization

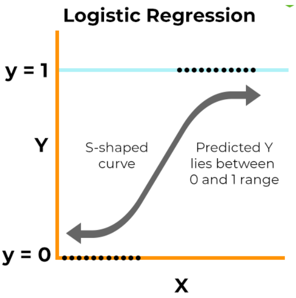

Figure 2: Logistic Regression Illustration [?]

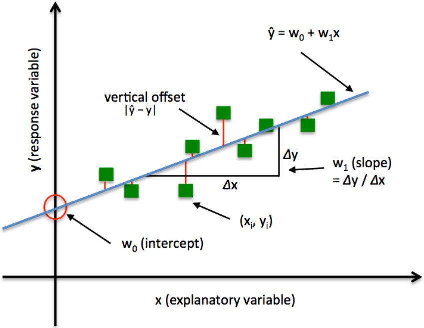

Figure 3: Linear Regression Illustration[?]

Monte Carlo Simulation

- Computational technique that uses repeated random sampling to estimate probabilities for systems with uncertainty.

- Generates N independent samples $x_1, x_2, \ldots, x_N$ from a distribution $f(x)$, then estimates probabilities by counting how often a condition is met:

$$\hat{P} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}(x_i \in A) \tag{4}$$

- $\mathbf{1}(\cdot)$: indicator function (1 if condition met, 0 otherwise).
- A: event of interest; $\hat{P}$: estimate of true probability $P(A)$.

- By the Law of Large Numbers [**?**], $\hat{P} \to P(A)$ as $N \to \infty$.

XGBoost: Overview

- **Extreme Gradient Boosting** [?] is an ensemble of decision trees built sequentially.
- Each new tree corrects the residual errors of the previous trees.
- Final prediction is the sum of contributions from all trees:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i) \tag{5}$$

- Widely used in competitive ML for structured data; captures complex interactions and nonlinearity automatically.

Building Each Tree

- Start with an initial prediction (e.g., the average outcome).
- Compute the **loss** (binary cross-entropy):

$$\ell(y_i, \hat{y}_i) = -[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{6}$$

- Compute the **gradient** of the loss with respect to each prediction:

$$g_i = \hat{y}_i - y_i \tag{7}$$

- The gradient tells each prediction how to move to reduce loss.
- The next tree is trained to predict these gradients across the feature space.

Objective Function

- Each tree is added to minimize the objective function:

$$\mathrm{obj}(\theta) = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{8}$$

- **Loss term** $\sum \ell(y_i, \hat{y}_i)$: total prediction error; the gradient is its derivative.
- **Regularization term** $\sum \Omega(f_k)$: penalizes complex trees:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{9}$$

- T: number of leaves; w_j: leaf values; γ, λ: regularization hyperparameters.
- Discourages overly deep trees and extreme leaf values.

XGBoost Example

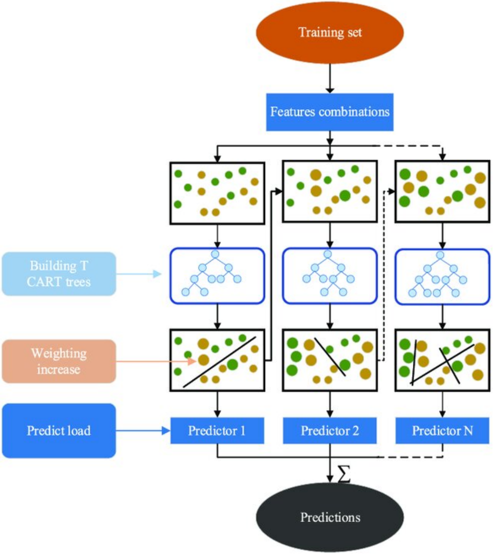

Figure 4: How XGBoost combines trees sequentially [?]

Cross Validation

- Evaluation technique that systematically rotates which portion of the data is held out, so every observation serves as both training and validation [**?**].
- In k-**fold cross validation**: partition data into k equally sized folds. Train k times, each time using $k-1$ folds for training and the remaining fold for validation.
- Final score is the average performance across all k folds.

CV Example

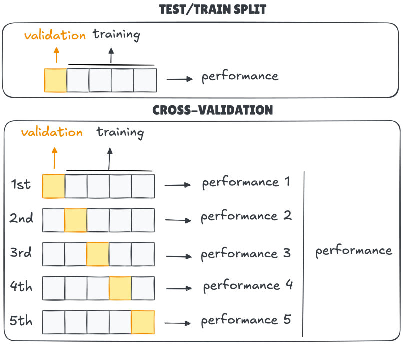

Figure 5: Cross Validation vs. Regular Train/Test Split [?]

Section 5

Methods

Logistic Regression: Setup

- Outcome is binary (win/loss), and logistic regression directly models the log-odds as a linear combination of features.
- Output is a calibrated probability so no transformation needed.
- Coefficients are interpretable, providing direct insight into which features drive predictions.
- **Training boundary:** features computed from regular season data, model trained on tournament outcomes.

Training Window

- To prevent data leakage and reflect roster turnover, the model uses a **rolling two-year window**.
- Example: to forecast the 2025 tournament, train only on 2023 and 2024 tournament games.
- **Motivation:** high roster turnover in college basketball means a program undergoes near-complete personnel transformation within 2–3 seasons.
- Older data does not reflect current team strength.

Variable Selection: The Pace Problem

- Raw box score totals are confounded by **pace**.
- High-tempo teams accumulate more counting stats without being more efficient.
- To compare teams fairly, statistics must be normalized by possessions.
- Solution: Dean Oliver's **Four Factors** [**?**], the components most correlated with winning.

The Four Factors

- Possession estimation from box score data:

$$\text{Poss} = (FGA - OR) + TO + 0.44 \cdot FTA \tag{10}$$

$$eFG\% = \frac{FGM + 0.5 \cdot FGM3}{FGA} \qquad TOV\% = \frac{TO}{\text{Poss}} \tag{11}$$

$$ORB\% = \frac{OR}{OR + DR_{opp}} \qquad FTR = \frac{FTM}{FGA} \tag{12}$$

- **Effective FG%**: adjusts for the added value of three-pointers.
- **Turnover %**: turnovers per possession.
- **Offensive Rebound %**: rebounding dominance per opportunity.
- **Free Throw Rate**: free throw frequency relative to FG attempts.

Final Feature Set

- Beyond the Four Factors, the model adds **Win Percentage** as a fifth feature.
- Win% aggregates effects the Four Factors miss directly:
  - Coaching decisions
  - Late-game execution
  - Opponent quality
- All five features computed as season averages from regular season data.
- Transformed into **differentials** between competing teams, centering the analysis on the relative mismatch.

Model Implementation

- Logistic regression implemented separately for Men's and Women's divisions.
- Why not one combined model with a gender indicator?
  - Differences in pace, physicality, and three-point volume.
  - The Four Factors carry structurally different predictive weight across divisions.
  - Forcing shared parameters would mask these differences.
- Each model optimizes its coefficients to the unique statistical footprint of its division.

Model Specification

- Logit link function maps linear combination of differentials to a probability in $(0, 1)$:

$$\text{logit}(\hat{P}) = \ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1(\text{eFG\_diff}) + \beta_2(\text{TOV\_diff}) + \beta_3(\text{ORB\_diff}) + \beta_4(\text{FTR\_diff}) + \beta_5(\text{Win\_diff}) \tag{13}$$

- P: probability of a Team 1 victory.
- Coefficients β estimated via maximum likelihood:

$$\sum_{i=1}^{n} \left[y_i \log p(x_i) + (1-y_i)\log(1-p(x_i))\right] \tag{14}$$

Multicollinearity Check

- All five predictors fall below VIF $= 3$ in both division so no problematic collinearity.

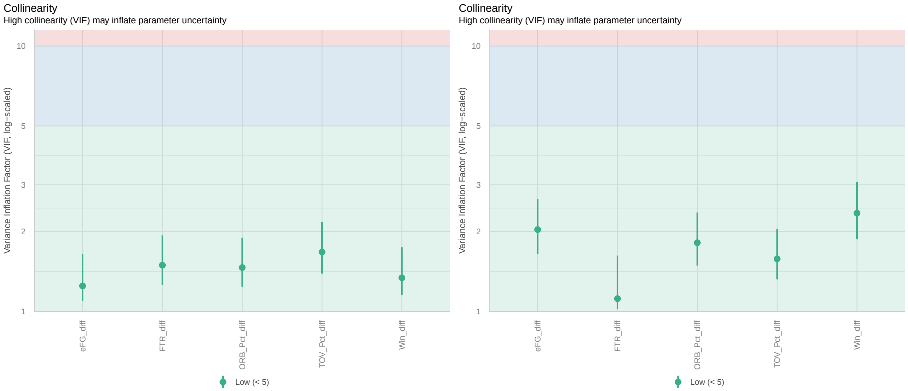

Figure 6: Men's (left) and Women's (right) VIF Plots for 2025

Feature Correlations

- Pairwise correlations are modest in both divisions; consistent with VIF results.

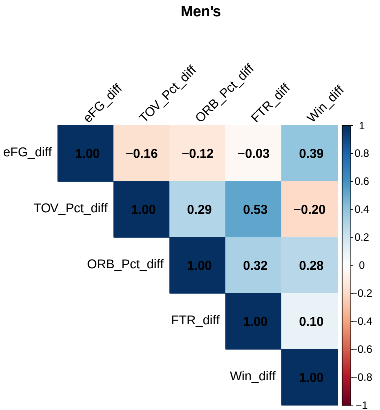

Figure 7: Men's (left) and Women's (right) Correlation Matrix for 2025

Coefficient Estimates

Men's Model

Variable	Est.	SE	p	Sig.
Intercept	0.4352	0.2062	0.0348	*
eFG Diff	7.7952	6.4160	0.2244	
TOV Pct Diff	-20.5106	12.1028	0.0901	.
ORB Pct Diff	12.3634	4.1219	0.0027	**
FTR Diff	-1.4307	4.5405	0.7527	
Win Diff	1.3253	1.7033	0.4365	

Women's Model

Variable	Est.	SE	p	Sig.
Intercept	-0.0888	0.2055	0.6657	
eFG Diff	17.4458	5.5413	0.0016	**
TOV Pct Diff	-19.8212	8.7655	0.0237	*
ORB Pct Diff	12.4308	3.5127	0.0004	***
FTR Diff	3.9742	5.1691	0.4420	
Win Diff	-1.8468	2.2764	0.4172	

- ORB% drives wins in both divisions (~ 12.4) — universal driver.
- eFG% significant in women's, not men's.